Portfolio Selection over Ordinal Uncertainty Sets*

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Abstract

Standard mean-variance optimization is fundamentally fragile to estimation errors in expected returns. Because the mapping from expected returns to portfolio weights is ill-conditioned, small cardinal errors in input forecasts frequently result in unbounded utility losses out-of-sample. This paper proposes an optimization framework that addresses this fragility by constraining the investor's uncertainty set to include only ordinal information. We replace precise point estimates of expected returns with an "Ordinal Uncertainty Set"—a polyhedral cone containing all return vectors that preserve the rank and sign of the initial forecast. By maximizing worst-case utility over this set, we derive a portfolio that mitigates cardinal errors while making full use of the investor's directional insights. We show analytically that this "ordinal regularization" provides a strict bound on worst-case utility loss, offering a theoretical justification for trading strategies based on ranking signals (such as analyst recommendation, momentum, and value scores) rather than precise return forecasts.

1 Introduction

The theoretical elegance of the Mean-Variance (MV) optimization framework introduced by Markowitz (1952) provides the cornerstone of modern portfolio theory. By formalizing the trade-off between expected return and risk, it offers a simple methodology for optimal asset allocation. However, the practical application of this framework has been problematic for decades. The MV optimizer, in its "plug-in" form, treats its inputs—the vector of expected returns ($\hat{\mu}$) and the covariance matrix (Σ)—as if they were known with perfect certainty. In practice, these parameters are often estimated from noisy historical data, and the resulting optimal portfolios are notoriously unstable and perform poorly out-of-sample (Michaud, 1989; Jorion, 1992).

The core of the issue lies in the fact that the MV optimization is an ill-conditioned problem. Michaud (1989) characterized the MV optimizer as an "error maximization machine," noting that it over-weights assets with large positive estimation errors in expected returns and under-weights

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(or shorts) those with large negative errors. Best and Grauer (1991) provided an analytical proof of this fragility, demonstrating that even minuscule, statistically insignificant changes in the expected return vector can lead to massive, economically significant changes in the resulting portfolio weights. This extreme sensitivity to inputs, particularly to $\hat{\mu}$, is not merely a statistical issue; it is a fundamental flaw that limits the practical reliability of the standard MV portfolio without significant modification.

In response to this fundamental fragility, this paper proposes an optimization framework that departs entirely from cardinal estimation. We argue that the root of the problem is the reliance on precise numerical forecasts, a requirement that is at odds with the true nature of most investment insights, which are often purely ordinal (e.g., rankings or signs). We therefore develop a method to optimize portfolios using only this ordinal information, introducing an "Ordinal Uncertainty Set" that limits the cardinal errors that affect standard optimization models. This approach provides a foundation for investment strategies based on directional views, a common practice that has, to date, lacked a formal optimization framework. Our central contribution is the introduction of an ordinal uncertainty set that preserves only the rank and sign of expected returns while allowing arbitrary cardinal noise, leading to a robust portfolio rule that endogenously enforces sign-consistency (justifying institutional long-only mandates) and is not nested within existing shrinkage or robust optimization frameworks.

The academic and practitioner response to this "cardinal fragility" has been vast, largely branching into two distinct streams. The first stream attempts to improve the quality of the cardinal inputs themselves. Bayesian methods, most notably the Black and Litterman (1992) model, provide a mechanism to blend an investor's subjective "views" with a market-implied equilibrium prior, effectively shrinking the expected return vector towards a more stable central tendency. In a similar vein, statistical shrinkage estimators, originating from the work of James and Stein (1992) and applied to finance by authors like Jorion (1986), pull extreme return estimates towards a "grand mean" to reduce the impact of estimation error. Other approaches rely on factor models to impose more structure on the problem (e.g., Fama and French, 1993) or use machine learning techniques to generate more stable forecasts. However, while these methods dampen estimation noise, they do not eliminate the fundamental reliance on cardinal magnitudes. They typically require the investor to specify precise hyperparameters, such as confidence levels or shrinkage intensities, that are themselves difficult to estimate, effectively replacing one cardinal estimation problem with another.

The second stream of literature, which more closely relates to our work, accepts the inputs as

inherently noisy and instead modifies the optimization problem to make it more robust. One of the simpler methodologies in this literature is the imposition of portfolio constraints. Jagannathan and Ma (2003) showed that imposing seemingly "wrong" constraints, such as a no-short-selling constraint, is mathematically equivalent to shrinking the covariance matrix, effectively regularizing the problem and often improving out-of-sample performance.

Another strand of the literature is related to early work by Litzenberger et al. (1971), who showed that simple ordinal predictions about common stocks contain useful information for portfolio selection, and to more recent approaches that integrate ordinal views into Black–Litterman-type estimators of expected returns (Çela et al., 2021, 2025). In contrast to these contributions, which ultimately map ordinal information back into a cardinal expected return vector, the framework developed here treats ordinal beliefs as primitive and defines portfolio choice as a robust optimization problem over the entire set of cardinal return vectors consistent with those beliefs. The resulting "ordinal uncertainty set" is a polyhedral cone, which allows a full geometric and dual characterization and yields explicit bounds on the regret of the standard mean–variance portfolio.

A more formal approach within this second stream is "Robust Optimization" (RO), pioneered by Ben-Tal and Nemirovski (1998). Instead of using a single point estimate $\hat{\mu}$, the RO framework assumes the true, unknown μ^* lies within a specified uncertainty set $\mathcal{U}$ centered around $\hat{\mu}$. The investor then solves a max-min problem, maximizing utility against the worst-case realization of μ within $\mathcal{U}$. However, the vast majority of the RO literature in finance defines these uncertainty sets using cardinal bounds. For example, Goldfarb and Iyengar (2003) propose ellipsoidal uncertainty sets (e.g., $(\mu - \hat{\mu})^T \Sigma^{-1} (\mu - \hat{\mu}) \leq \delta$), which are statistically motivated by the confidence region of $\hat{\mu}$. Other common approaches use simple box constraints (e.g., $l_i \leq \mu_i \leq u_i$) (Tütüncü and Koenig, 2004; Fabozzi et al., 2007). While these methods provide protection against estimation error within the specified cardinal bounds, they share the same fundamental weakness as the original plug-in approach: their output is still highly sensitive to the initial point estimate $\hat{\mu}$ and the cardinal size (δ) of the set. An investor must still provide a precise "best guess" $\hat{\mu}$ and a precise "confidence level" δ, both of which are notoriously difficult to specify.

This paper argues that both of these streams—statistical and robust—fail to address the true nature of reliable investment information. In many, if not most, practical investment settings, an investor's "alpha" or "edge" is not cardinal; it is ordinal. An analyst may have high confidence that Asset A will outperform Asset B, or that a portfolio of "value" stocks will outperform "growth" stocks, without having any reliable basis for forecasting that the spread will be 2% or 8%. Similarly,

fundamental analysis or momentum signals are far more reliable at ranking assets than at producing precise return forecasts.

A large body of evidence in behavioral decision theory suggests that human judgments are fundamentally ordinal rather than cardinal. Individuals are typically consistent in ranking outcomes, yet unreliable in assessing magnitudes (Tversky, 1969; Gigerenzer and Goldstein, 1996). This asymmetry reflects a cognitive constraint: under uncertainty, agents manage complexity through categorization—mapping continuous variables into discrete buckets—rather than precise estimation (Mullainathan, 2002; Shue et al., 2024).

Similarly, the literature on rational inattention shows that individuals process information "coarsely," often exhibiting a "left-digit bias" where focus is placed on higher-order magnitude indicators (like sign or rank) while lower-order cardinal details are neglected (Gabaix, 2019). In forecasting, this manifests as higher reliability of directional predictions relative to level forecasts (Goodwin and Fildes, 1999). The framework proposed here formalizes this behavioral regularity: we define an optimization problem where the information set is treated as a coarse category (ordinal) rather than a precise value (cardinal), consistent with how beliefs are actually formed.

Existing frameworks are generally not well equipped to handle this type of ordinal setup. The Black-Litterman model, for instance, still requires the user to specify a cardinal expected return for their view (e.g., "Asset A will outperform B by 2%"). Standard robust optimization requires a cardinal anchor $\hat{\mu}$. These frameworks force the investor to invent cardinal precision where none exists, re-introducing the very estimation error they seek to avoid.

This paper develops an optimization framework that addresses this fragility by reformulating the uncertainty set to preserve only ordinal information—the ranking and signs of expected returns—while allowing unlimited cardinal error. We formally define an "Ordinal Uncertainty Set" as the set of all return vectors that are ordinally consistent with an investor's forecast. By maximizing worst-case utility over this set, we derive a portfolio that is immune (or at least limits) the possibility of cardinal errors while making full use of the investor's directional insights.

Our main contributions are threefold. First, we provide a mathematical characterization of the ordinal uncertainty set as a polyhedral cone and establish its geometric and algebraic properties. Second, we demonstrate, analytically and via numerical simulation, that the standard mean-variance portfolio suffers unbounded utility loss even when estimation errors preserve perfect ordinal accuracy, whereas our ordinal portfolio maintains bounded "regret." Third, we derive a computationally tractable reformulation of the optimization problem through convex duality and validate the frame-

work empirically, showing that our robust solution offers the same computational complexity as the standard plug-in approach but yields superior out-of-sample stability and diversification.

2 Model Setup and the Fragility of Cardinal Optimization

The foundational insight of modern portfolio theory (Markowitz, 1952) is that an investor's optimal portfolio is a trade-off between expected return and variance. The resulting Mean-Variance (MV) framework is elegant, intuitive, and theoretically important. However, its practical application has been notoriously problematic (Michaud, 1989; Best and Grauer, 1991; Markowitz and Todd, 2000).

Decades of empirical work have shown that the "optimal" portfolios generated by the standard MV "plug-in" approach are often incorrect, consisting of extreme long and short positions that are highly unstable and perform poorly out-of-sample. Michaud (1989) famously labeled the framework an "error maximization machine." The core problem is that the MV optimizer is hypersensitive to its inputs, particularly the vector of expected returns, μ. As Best and Grauer (1991) demonstrate, small, statistically insignificant changes in μ can lead to massive, economically significant changes in the resulting portfolio weights. This paper argues that this fragility is not just a statistical estimation problem but a fundamental mathematical one. The standard MV optimizer is designed to trust cardinal precision—it treats a forecast of 5.1% as infinitely different from 5.0%. This reliance on precise, cardinal inputs is the root of its instability.

Existing responses to this fragility have largely focused on "robustifying" cardinal inputs. Classical robust-optimization treatments of mean–variance uncertainty remain fundamentally cardinal in nature: they restrict the magnitudes of estimation errors in the expected return vector while preserving reliance on precise numerical forecasts. We depart from this paradigm by constructing an ordinal uncertainty set, in which the investor only trusts the sign and rank ordering of expected returns, while allowing for arbitrary cardinal noise. This structure is economically natural—ordinal beliefs are typically far more stable than cardinal ones—and mathematically consequential, because it induces a conical uncertainty set. As we show, this change in structure fundamentally alters the worst-case problem, yielding (i) bounded regret, (ii) a radically different optimal portfolio characterized by sign- and rank-monotonicity, and (iii) stability properties that are not shared by cardinal robust optimization, shrinkage, or Bayesian approaches.

To formally demonstrate this "cardinal fragility" and motivate our ordinal solution, we begin by establishing the standard MV framework. Consider a frictionless market with N risky assets.

Let $R \in \mathbb{R}^N$ be the random vector of asset excess returns, with the first two moments given by:

$$\begin{aligned} \mathbb{E}[R] &= \mu^* \\ \mathbb{V}[R] &= \Sigma \end{aligned} \tag{1}$$

where μ^* is the (unobservable) vector of true expected excess returns, and Σ is the true positive-definite covariance matrix. For the purposes of focusing on first-moment estimation risk, we assume Σ is known.[1]

A representative investor with risk aversion coefficient $\lambda > 0$ seeks to maximize standard mean-variance utility. If the true parameters μ^* were known, the investor would solve the following unconstrained problem:

$$\max_{w \in \mathbb{R}^N} U(w, \mu^*) = w^\top \mu^* - \frac{\lambda}{2} w^\top \Sigma w \tag{2}$$

The first-order condition $\mu^* - \lambda \Sigma w = 0$ yields the classic "true" optimal portfolio weights:

$$w^* = \frac{1}{\lambda} \Sigma^{-1} \mu^* \tag{3}$$

This vector w^* represents the theoretical ideal benchmark against which realized portfolio performance must be evaluated.

In practice, μ^* is unknown. The investor must rely on an estimator $\hat{\mu}$, which can be decomposed into the true signal plus an error term:

$$\hat{\mu} = \mu^* + \epsilon, \quad \epsilon \sim \mathcal{D}(0, \Omega_\epsilon) \tag{4}$$

where ϵ represents estimation noise. The standard "plug-in" approach treats this cardinal estimate as truth, resulting in the realized portfolio weights:

$$\hat{w}_{MV} = \frac{1}{\lambda} \Sigma^{-1} \hat{\mu} = \frac{1}{\lambda} \Sigma^{-1} (\mu^* + \epsilon) = w^* + \frac{1}{\lambda} \Sigma^{-1} \epsilon \tag{5}$$

The term $\frac{1}{\lambda} \Sigma^{-1} \epsilon$ represents the deviation from optimality. We now establish formally that this deviation can lead to unbounded utility loss.

Proposition 1 (Unbounded Utility Loss Under Cardinal Errors)**.** Let $\lambda_{\min}(\Sigma)$ denote the smallest

[1] This is a standard assumption in the robust portfolio literature to isolate the impact of errors in μ, which are empirically shown to be an order of magnitude more consequential than errors in Σ (Merton, 1980; Chopra et al., 1993).

eigenvalue of Σ. For any $\delta > 0$, there exists an error vector ϵ with $\|\epsilon\|_2 = \delta$ such that as $\lambda_{\min}(\Sigma) \to 0$:

$$U(w^*, \mu^*) - U(\hat{w}_{MV}, \mu^*) \to \infty \tag{6}$$

Proof. The utility loss (or "regret") is the difference between the oracle's utility and the plug-in portfolio's utility, $L = U(w^*, \mu^*) - U(\hat{w}_{MV}, \mu^*)$. A Taylor expansion of $U(w, \mu^*)$ around the optimum w^* shows the loss is exactly a quadratic form, as the first-order term is zero by definition of w^*:

$$L = \frac{\lambda}{2}(\hat{w}_{MV} - w^*)^\top \Sigma (\hat{w}_{MV} - w^*) \tag{7}$$

Let $\Delta w = \hat{w}_{MV} - w^* = \frac{1}{\lambda}\Sigma^{-1}\epsilon$. Substituting this into the loss function gives:

$$\begin{aligned} L &= \frac{\lambda}{2}\left(\frac{1}{\lambda}\Sigma^{-1}\epsilon\right)^\top \Sigma \left(\frac{1}{\lambda}\Sigma^{-1}\epsilon\right) \\ &= \frac{\lambda}{2}\frac{1}{\lambda^2}\epsilon^\top (\Sigma^{-1})^\top \Sigma \Sigma^{-1}\epsilon \\ &= \frac{1}{2\lambda}\epsilon^\top \Sigma^{-1}\epsilon \end{aligned} \tag{8}$$

Now, let $v_{\min}$ be the eigenvector corresponding to $\lambda_{\min}(\Sigma)$, normalized such that $\|v_{\min}\|_2 = 1$. Consider the adversarial error vector $\epsilon = \delta v_{\min}$ for any $\delta > 0$. The utility loss becomes:

$$\begin{aligned} L &= \frac{1}{2\lambda}(\delta v_{\min})^\top \Sigma^{-1}(\delta v_{\min}) \\ &= \frac{\delta^2}{2\lambda} v_{\min}^\top \Sigma^{-1} v_{\min} \end{aligned} \tag{9}$$

Since $\Sigma v_{\min} = \lambda_{\min}(\Sigma) v_{\min}$, it follows that $\Sigma^{-1} v_{\min} = \frac{1}{\lambda_{\min}(\Sigma)} v_{\min}$. Substituting this in yields:

$$\begin{aligned} L &= \frac{\delta^2}{2\lambda} v_{\min}^\top \left(\frac{1}{\lambda_{\min}(\Sigma)} v_{\min}\right) \\ &= \frac{\delta^2}{2\lambda\lambda_{\min}(\Sigma)}(v_{\min}^\top v_{\min}) \\ &= \frac{\delta^2}{2\lambda\lambda_{\min}(\Sigma)} \end{aligned} \tag{10}$$

As $\lambda_{\min}(\Sigma) \to 0$, the utility loss $L \to \infty$. This holds for any $\delta > 0$, proving that even infinitesimal cardinal errors can lead to unbounded utility loss if Σ is ill-conditioned. □

The result in Proposition 1 provides the mathematical insight for the "error maximization" critique. The unbounded utility loss is not just a theoretical construct; it arises under the most

common conditions in financial markets. The condition $\lambda_{\min}(\Sigma) \to 0$ represents an ill-conditioned covariance matrix, which occurs whenever assets are highly correlated or co-integrated—a common feature of real-world asset classes. The error vector ϵ is precisely an error in the direction of $v_{\min}$, the portfolio that combines these assets into a (perceived) near-arbitrage, low-variance position. The optimizer, in its blind trust of the cardinal inputs, misinterprets this estimation error ϵ as a genuine, low-risk return opportunity and builds an enormous, unstable position to exploit it. This is the exact mechanism of cardinal fragility.[2]

It is critical to note that this failure does not require the error ϵ to be "large" in norm (the proof holds for any $\delta > 0$). Nor does it require the error to be ordinally incorrect. A forecast $\hat{\mu} = \mu^* + \epsilon$ could, for instance, preserve the exact ranking and signs of the true returns μ^*, yet if the cardinal error ϵ aligns with $v_{\min}$, the resulting plug-in portfolio $\hat{w}_{MV}$ will still suffer large utility losses. This shows that the problem is not merely "estimation error" in a statistical sense, but rather the optimizer's fundamental reliance on cardinal precision. This failure motivates a departure from traditional portfolio optimization techniques: if the standard optimizer fails by trusting cardinal information, the solution must be a framework that explicitly discards this information, relying only on the more robust, ordinal properties of the forecast. This leads us directly to the formulation of an uncertainty set that captures this logic.

3 Ordinal Uncertainty Sets

To address the fragility of cardinal optimization, we propose a shift in how input estimates are treated. Rather than attempting to "fix" the estimate $\hat{\mu}$ through statistical shrinkage—which still results in a cardinal point estimate subject to the same failure modes—we redefine the inputs to the optimization problem. This section formally defines the central object of our analysis, the Ordinal Uncertainty Set, by building it from its two constituent components: sign and rank.

Let $\hat{\mu} \in \mathbb{R}^N$ be the initial noisy forecast. We define ordinal consistency through two fundamental properties:

Definition 1 (Sign Preservation)**.** A vector $\mu \in \mathbb{R}^N$ preserves the sign of $\hat{\mu}$ if, for all assets $i \in$

[2] Practitioners often impose ad hoc weight constraints (e.g., $|w_i| \leq 5\%$) to mechanically prevent such extreme positions. While this technically precludes unbounded loss, the choice of such bounds is often arbitrary and does not resolve the underlying inference problem. Instead of filtering the noise, such constraints merely truncate the optimizer's reaction to it, often forcing the portfolio to be determined by the boundaries of the feasible set rather than the information in the signal.

$\{1, \ldots, N\}$:

$$\text{sgn}(\mu_i) = \text{sgn}(\hat{\mu}_i) \tag{11}$$

where $\text{sgn}(x) = 1$ if $x > 0$, $\text{sgn}(x) = -1$ if $x < 0$, and $\text{sgn}(0) = 0$.

Definition 1 formalizes the investor's most basic directional view. It partitions the asset universe into three distinct sets: expected winners (potential longs), expected losers (potential shorts or underweights), and neutral assets. This constraint is the mathematical formulation of a 'buy-versus-sell' signal. It ensures that our framework respects this fundamental premise; for example, the worst-case scenario for a 'buy' signal ($\hat{\mu}_i > 0$) cannot be a reality where that asset actually has a negative expected return ($\mu_i < 0$).

Definition 2 (Rank Preservation)**.** Let $\pi : \{1, \ldots, N\} \to \{1, \ldots, N\}$ be a permutation that sorts the elements of $\hat{\mu}$ in descending order, such that $\hat{\mu}_{\pi(1)} \geq \hat{\mu}_{\pi(2)} \geq \cdots \geq \hat{\mu}_{\pi(N)}$. A vector μ preserves the rank of $\hat{\mu}$ if:

$$\mu_{\pi(1)} \geq \mu_{\pi(2)} \geq \cdots \geq \mu_{\pi(N)} \tag{12}$$

Definition 2 introduces a much richer, more granular set of beliefs and is the mathematical embodiment of 'relative value'. It represents the common investment case where an analyst is far more confident in their ability to rank a set of assets (e.g., "we like Asset A more than Asset B, and B more than C") than in their ability to assign a precise cardinal forecast to any of them. This is the core information generated by most screening models (e.g., momentum, value factors) and qualitative research, which has historically been difficult to incorporate into a formal optimization.

Combining Definitions 1 and 2, we obtain the set of all return vectors that are consistent with both the investor's directional and relative-ordering beliefs. This, then, allows us to define the central object of our analysis by combining these two concepts:

Definition 3 (Ordinal Uncertainty Set)**.** Given a forecast $\hat{\mu} \in \mathbb{R}^N$ with no zero components, the ordinal uncertainty set is:

$$\mathcal{U}_{ord}(\hat{\mu}) = \left\{ \mu \in \mathbb{R}^N \;\middle|\; \begin{array}{l} \text{sgn}(\mu_i) = \text{sgn}(\hat{\mu}_i), \quad \forall i \in \{1, \ldots, N\} \\ \mu_{\pi(1)} \geq \mu_{\pi(2)} \geq \cdots \geq \mu_{\pi(N)} \end{array} \right\} \tag{13}$$

The Ordinal Uncertainty Set $\mathcal{U}_{ord}(\hat{\mu})$ is the geometric representation of the "universe of all possible realities" that are consistent with the investor's non-cardinal information. Any vector μ in this set is 'ordinally consistent' with the forecast $\hat{\mu}$, but the cardinal magnitudes of the returns and

the spreads between them are unconstrained. For example, if the forecast is $\hat{\mu} = (5\%, 2\%, -1\%)^T$, this set contains both a favorable reality like $\mu = (10\%, 9\%, -0.5\%)^T$ and an adversarial, "worst-case" reality like $\mu = (0.01\%, 0.001\%, -20\%)^T$. By defining this set, we can now formulate a max-min problem: how to build a portfolio that performs best, given that an adversary will pick the single worst μ from this entire set of ordinally-consistent realities.

We now establish the key geometric properties of this set:

Theorem 1 (Geometric Characterization)**.** The ordinal uncertainty set $\mathcal{U}_{ord}(\hat{\mu})$ is a convex polyhedral cone with the following properties:

a) **Convexity:** For any $\mu_1, \mu_2 \in \mathcal{U}_{ord}(\hat{\mu})$ and $\alpha \in [0, 1]$:

$$\alpha\mu_1 + (1 - \alpha)\mu_2 \in \mathcal{U}_{ord}(\hat{\mu}) \tag{14}$$

b) **Conicity:** For any $\mu \in \mathcal{U}_{ord}(\hat{\mu})$ and $t > 0$:

$$t\mu \in \mathcal{U}_{ord}(\hat{\mu}) \tag{15}$$

c) **Polyhedral Structure:** $\mathcal{U}_{ord}(\hat{\mu})$ can be represented as:

$$\mathcal{U}_{ord}(\hat{\mu}) = \{\mu \in \mathbb{R}^N : A\mu \geq 0\} \tag{16}$$

where $A \in \mathbb{R}^{m \times N}$ is a matrix encoding the sign and rank constraints, with $m = N + (N - 1)$.

d) **Non-emptiness:** $\hat{\mu} \in \mathcal{U}_{ord}(\hat{\mu})$ by construction.

These four properties are the key theoretical pillars that make our framework viable. Convexity (a) ensures that the robust problem is well-posed and avoids the intractability of non-convex optimization. Conicity (b) is the direct mathematical expression of "unlimited cardinal error"—the defining feature of our set. Finally, the Polyhedral Structure (c) is the important component for computation, as it proves that our set, despite its financial motivation, is a standard geometric object that can be represented by simple linear inequalities. This property is what ultimately allows us to reformulate the problem into a tractable, efficient program.

Proof. **Part (a) Convexity:** Let $\mu_1, \mu_2 \in \mathcal{U}_{ord}(\hat{\mu})$ and $\alpha \in [0, 1]$. Define $\mu = \alpha\mu_1 + (1 - \alpha)\mu_2$. For sign preservation: Since $\text{sgn}(\mu_{1,i}) = \text{sgn}(\mu_{2,i}) = \text{sgn}(\hat{\mu}_i)$ for all i, and both $\alpha \geq 0$ and $(1 - \alpha) \geq 0$,

μ_i must have the same sign as $\mu_{1,i}$ and $\mu_{2,i}$.

$$\text{sgn}(\mu_i) = \text{sgn}(\alpha\mu_{1,i} + (1-\alpha)\mu_{2,i}) = \text{sgn}(\hat{\mu}_i) \tag{17}$$

For rank preservation: Since $\mu_{1,\pi(j)} \geq \mu_{1,\pi(j+1)}$ and $\mu_{2,\pi(j)} \geq \mu_{2,\pi(j+1)}$ for all j:

$$\begin{aligned} \mu_{\pi(j)} &= \alpha\mu_{1,\pi(j)} + (1-\alpha)\mu_{2,\pi(j)} \\ &\geq \alpha\mu_{1,\pi(j+1)} + (1-\alpha)\mu_{2,\pi(j+1)} = \mu_{\pi(j+1)} \end{aligned} \tag{18}$$

Thus, $\mu \in \mathcal{U}_{ord}(\hat{\mu})$.

Part (b) Conicity: For $\mu \in \mathcal{U}_{ord}(\hat{\mu})$ and $t > 0$:

- Sign preservation: $\text{sgn}(t\mu_i) = \text{sgn}(\mu_i) = \text{sgn}(\hat{\mu}_i)$ since $t > 0$.
- Rank preservation: $\mu_{\pi(j)} \geq \mu_{\pi(j+1)} \implies t\mu_{\pi(j)} \geq t\mu_{\pi(j+1)}$ since $t > 0$.

Thus, $t\mu \in \mathcal{U}_{ord}(\hat{\mu})$.

Part (c) Polyhedral Structure: Define the constraint matrix $A \in \mathbb{R}^{(2N-1)\times N}$ with rows $a_k^\top$:

- For N sign constraints ($k = 1, \ldots, N$): $a_k^\top = e_k^\top \cdot \text{sgn}(\hat{\mu}_k)$
- For $N-1$ rank constraints ($k = N+1, \ldots, 2N-1$): $a_k^\top = e_{\pi(j)}^\top - e_{\pi(j+1)}^\top$, where $j = k - N$.

where e_i is the i-th standard basis vector. The set of linear inequalities $\mu_i \cdot \text{sgn}(\hat{\mu}_i) \geq 0$ (which defines the sign) and $\mu_{\pi(j)} - \mu_{\pi(j+1)} \geq 0$ (which defines the rank) is exactly equivalent to the system $A\mu \geq 0$. The set is therefore a polyhedral cone.

Part (d) Non-emptiness: Clearly $\hat{\mu} \in \mathcal{U}_{ord}(\hat{\mu})$ by definition, so the set is non-empty. □

To understand the structure of the ordinal uncertainty set, we turn to characterizing its "edges" or "corners."

Proposition 2 (Extreme Rays)**.** The extreme rays of $\mathcal{U}_{ord}(\hat{\mu})$ are vectors $r \in \mathbb{R}^N$ generated by $N-1$ linearly independent, active constraints from A. A non-trivial extreme ray corresponds to a reality where all assets within a given sign group have the same expected return.

Proof. By the theory of polyhedral cones (e.g., Rockafellar, 1970), the extreme rays are the one-dimensional nullspaces formed by the intersection of $N-1$ linearly independent, active constraints from the matrix A. An active rank constraint $a_k^\top \mu = 0$ implies $\mu_{\pi(j)} = \mu_{\pi(j+1)}$. An active sign

constraint $a_k^\top \mu = 0$ implies $\mu_k = 0$. A non-trivial ray (i.e., not the zero vector) will be formed by $N-1$ active rank constraints, which implies $\mu_{\pi(1)} = \cdots = \mu_{\pi(k)}$ and $\mu_{\pi(k+1)} = \cdots = \mu_{\pi(N)}$ (where the split occurs at the sign change), thus "flattening" the expected returns within each sign-group. □

Proposition 2 gives financial intuition for the "adversary" in our robust problem. A well-known result in robust optimization is that the worst-case scenario over a polyhedron must occur at an extreme point (or, for a cone, on an extreme ray) (Chazelle, 1984). This proposition tells us exactly what the adversary's strategy is: to eliminate all spreads within our ranked assets. For an investor with a forecast $\hat{\mu} = (5\%, 3\%, 1\%)^T$, the adversary's "worst-case" μ is not $(5\%, 3\%, 1\%)$, but rather a reality like $(c, c, c)^T$ for some $c \to 0^+$. In this scenario, all the investor's relative-value insights are "flattened" to zero, and all assets are perceived to have the same minimal positive return. The robust optimizer is therefore forced to build a portfolio that is resilient to this "great flattening"—a portfolio that does not over-rely on perceived spreads that may be illusory.

4 The Robust Ordinal Portfolio

Having characterized the ordinal uncertainty set $\mathcal{U}_{ord}(\hat{\mu})$ and established its fundamental geometric properties in Section 3, we now use this set as the foundation for our robust portfolio selection model. The problem transitions from defining the boundaries of our belief (the set) to finding the optimal action to take given that our belief is inherently incomplete (the optimization).

The robust ordinal portfolio w_{rob} is the solution to a max-min game between the investor and an adversary who represents the market's unbounded cardinal uncertainty. The investor chooses a portfolio w, and the adversary, knowing w, chooses the worst-case expected return vector μ from within the investor's own uncertainty set. The investor's optimal strategy is the portfolio that performs best under this worst-case scenario. This is formally stated as:

$$\max_{w \in \mathcal{W}} \min_{\mu \in \mathcal{U}_{ord}(\hat{\mu})} U(w, \mu) = \max_{w \in \mathcal{W}} \left\{ \min_{\mu \in \mathcal{U}_{ord}(\hat{\mu})} (w^\top \mu) - \frac{\lambda}{2} w^\top \Sigma w \right\} \tag{19}$$

where $\mathcal{W}$ represents the set of permissible portfolios, such as the unconstrained case ($\mathcal{W} = \mathbb{R}^N$) or the fully-invested, budget-constrained case ($\mathcal{W} = \{w \in \mathbb{R}^N : w^\top \mathbf{1} = 1\}$).

It is instructive to parse the objective function in Equation 19. The risk term, $-\frac{\lambda}{2} w^\top \Sigma w$, is not subject to the inner minimization because we have assumed Σ is known and deterministic. The ad-

versary's power is limited to controlling the realization of μ. The inner problem, $\min_{\mu \in \mathcal{U}_{ord}(\hat{\mu})}(w^\top \mu)$, represents the "adversarial" or "worst-case" expected return for a given portfolio w. This adversary is the mathematical embodiment of cardinal uncertainty; they will "stress test" the portfolio by finding the single vector of expected returns μ, from the entire universe of ordinally-consistent realities, that is most damaging to w. The outer problem, $\max_{w \in \mathcal{W}}\{\dots\}$, represents the investor's response. The investor, knowing their portfolio will be subjected to this stress test, chooses the portfolio w that has the best worst-case outcome. This framework thus generates a portfolio w_{rob} that is, by construction, resilient to any possible cardinal realization of returns, so long as the investor's original ordinal forecast holds true.

Before solving this problem, we must first establish the conditions under which a solution exists. The conical (unbounded) nature of $\mathcal{U}_{ord}(\hat{\mu})$ means that the inner minimization problem is not guaranteed to be bounded. We establish a simple, intuitive condition on w that ensures boundedness.

Theorem 2 (Sign Consistency Condition)**.** For the inner minimization problem in (19) to be bounded below (i.e., for the worst-case utility to be finite), the portfolio weights must satisfy:

$$w_i \cdot \hat{\mu}_i \geq 0, \quad \forall i \in \{1, \dots, N\} \tag{20}$$

That is, the portfolio must be "sign-consistent" with the forecast: $\text{sgn}(w_i) = \text{sgn}(\hat{\mu}_i)$ or $w_i = 0$.

Proof. Suppose there exists an index i such that $w_i \cdot \hat{\mu}_i < 0$ (i.e., w_i and $\hat{\mu}_i$ have opposite signs).

- **Case 1: $w_i > 0$ and $\hat{\mu}_i < 0$.** The investor is long an asset they believe to be a "loser." By Definition 3, any $\mu \in \mathcal{U}_{ord}(\hat{\mu})$ must have $\mu_i < 0$. By the conicity property (Theorem 1), $t\mu \in \mathcal{U}_{ord}(\hat{\mu})$ for any $t > 0$. The adversary can thus construct a sequence of realities $\mu^{(k)}$ (e.g., by letting $\mu_i = -k$ for $k \to \infty$ while holding other elements constant in an ordinally-consistent way) such that the portfolio's expected return $w^\top \mu^{(k)} = w_i \cdot (-k) + \sum_{j \neq i} w_j \mu_j \to -\infty$.

- **Case 2: $w_i < 0$ and $\hat{\mu}_i > 0$.** The investor is short an asset they believe to be a "winner." By the same logic, any $\mu \in \mathcal{U}_{ord}(\hat{\mu})$ must have $\mu_i > 0$. The adversary can pick a sequence $\mu^{(k)}$ with $\mu_i = +k$ for $k \to \infty$, causing $w^\top \mu^{(k)} = w_i \cdot (+k) + \sum_{j \neq i} w_j \mu_j \to -\infty$.

In both cases, a sign-inconsistent portfolio suffers unbounded negative utility in the worst case. Therefore, for bounded utility (i.e., for the problem to be feasible), we must require $w_i \cdot \hat{\mu}_i \geq 0$ for all i. □

This theorem provides an important and intuitive financial justification for what is often an ad hoc constraint. The "Sign Consistency Condition" is not an arbitrary rule imposed by the investor; it is an endogenous and necessary result of accepting "unlimited cardinal error." It provides an answer to the question: "Why not short a high-momentum stock?" The standard cardinal answer is "because you might lose a few percent." Our ordinal answer is "because if you are wrong about the sign, your loss is unbounded." By embracing ordinality, the model itself derives this common-sense rule: one cannot take a portfolio position that directly "fights" the sign of the forecast, as doing so creates a form of model-based arbitrage for the adversary, who can inflict infinite losses. This result is the first key mechanism of "ordinal regularization"—it restricts the portfolio w to a feasible subspace that respects the investor's most basic directional views.

A mathematical feature of the pure ordinal approach, however, is that all dominance relations are treated uniformly: the constraint "Asset A is expected to outperform B" is encoded identically whether the underlying cardinal difference is economically small or large, as long as the sign is correct. Consequently, even very slight ordinal distinctions (e.g., $\mu_A \geq \mu_B$) can induce non-trivial tilts in the robust portfolio, because the worst-case analysis considers all cardinal realizations consistent with that ordering. This is not a defect of the model but an inherent implication of representing beliefs solely through orderings with unbounded cardinal uncertainty. In practice, the framework is most appropriate when the investor's directional views reflect economically meaningful distinctions rather than near-ties between substitutable assets.

The max-min problem in (19), coupled with the sign-consistency condition in (20), defines the robust portfolio. However, in this form, it is computationally opaque. To make the problem tractable, we must convert this nested, bi-level problem into a single, standard optimization problem. We achieve this by applying linear programming duality to the inner minimization problem, which is possible precisely because $\mathcal{U}_{ord}(\hat{\mu})$ is a polyhedral cone (Theorem 1).

Theorem 3 (Dual Reformulation)**.** The robust optimization problem (19), restricted to the feasible set of sign-consistent portfolios, is equivalent to the following convex Quadratic Program (QP):

$$\max_{w,\nu} \left\{ -\frac{\lambda}{2} w^\top \Sigma w \right\} \tag{21}$$

subject to:

$$
\begin{aligned}
w - A^\top \nu &= \mathbf{0} \\
\nu &\geq 0 \\
w &\in \mathcal{W} \\
w_i \cdot \hat{\mu}_i &\geq 0, \quad \forall i
\end{aligned}
$$

where A is the $m \times N$ constraint matrix from Theorem 1 and $\nu \in \mathbb{R}^m$ are the dual variables corresponding to the $m = 2N - 1$ ordinal constraints.

Proof. Fix a sign-consistent $w \in \mathcal{W}$. The inner minimization is a Linear Program (LP):

$$
\min_{\mu \in \mathcal{U}_{ord}(\hat{\mu})} w^\top \mu = \min_{\mu : A\mu \geq 0} w^\top \mu \tag{22}
$$

By strong duality (e.g., Rockafellar, 1970), this is equivalent to its dual LP:

$$
\max_{\nu \geq 0} \min_{\mu} \left\{ w^\top \mu - \nu^\top (A\mu) \right\} = \max_{\nu \geq 0} \min_{\mu} \left\{ (w - A^\top \nu)^\top \mu \right\} \tag{23}
$$

The inner unconstrained minimization over μ yields a value of 0 if $w - A^\top \nu = \mathbf{0}$, and $-\infty$ otherwise. To maximize, the dual variables ν must therefore be chosen to satisfy $w = A^\top \nu$ and $\nu \geq 0$. If such a ν exists, the worst-case expected return is 0. If no such $\nu \geq 0$ exists, the inner problem is unbounded (which is ruled out by Theorem 2, but the dual confirms this).

Substituting this result back into the outer problem (19), the investor's problem becomes:

$$
\max_{w \in \mathcal{W}} \left\{ 0 - \frac{\lambda}{2} w^\top \Sigma w \right\} \quad \text{s.t.} \quad w = A^\top \nu, \ \nu \geq 0 \tag{24}
$$

We can combine the investor's choice variable w and the dual variable ν into a single, unified optimization problem. The sign-consistency condition, $w_i \cdot \hat{\mu}_i \geq 0$, is added to explicitly enforce the feasible set for w derived in Theorem 2, leading to the formulation in (21). □

This dual reformulation is the central operational result of the paper. It transforms the abstract, "unsolvable" max-min game into a standard, convex Quadratic Program. This is an important practical finding: it proves that our ordinally robust portfolio is no harder to compute than the original, fragile Markowitz portfolio, as both are standard QPs.

The constraint $w = A^\top \nu$ has an important economic interpretation. Recall that the rows of A represent the sign and rank constraints. This equation shows that the optimal robust portfolio w must be a non-negative linear combination of these constraint vectors. The dual variables $\nu \geq 0$ act as the "shadow prices" or "importance" that the optimizer endogenously assigns to each ordinal constraint. The optimizer "builds" the portfolio w by "buying" (i.e., weighting by ν_k) the specific sign and rank constraints that are most effective at generating robust, risk-adjusted returns. For example, if the rank constraint "Asset A > Asset B" is particularly valuable (i.e., it holds across high-risk states), the optimizer will assign a large ν_k to it, and that constraint vector $(e_A - e_B)$ will become a key component of the final portfolio.

Corollary 1 (Endogenous Sparsity)**.** When input signals are positive ($\hat{\mu} > 0$), the Sign Consistency Condition (Theorem 2) requires that the robust ordinal portfolio be strictly long-only ($w \geq 0$). Furthermore, the solution is naturally sparse: the optimizer assigns non-zero weight to an asset only if its ordinal superiority guarantees a marginal utility improvement under the worst-case realization.

This result highlights a fundamental distinction between cardinal and ordinal optimization. In the standard mean–variance framework, any cardinal difference in expected returns—no matter how statistically insignificant—forces a reallocation of capital. If $\hat{\mu}_A > \hat{\mu}_B$, the standard optimizer will overweight A relative to B even if the difference is infinitesimal, leading to dense, unstable portfolios.

In the ordinal framework, dominance is necessary but not sufficient for capital allocation. While the uncertainty set $\mathcal{U}_{ord}$ enforces the ranking $\mu_A \geq \mu_B$, it also contains adversarial realizations where the spread between ranks collapses to zero (i.e., $\mu_A = \mu_B$). If the optimizer cannot certify that Asset A provides a strictly higher worst-case contribution than Asset B, it treats them as effectively equivalent in the worst case. Consequently, the robust portfolio exhibits "action thresholds": it ignores "noise" in the rankings and concentrates capital only on assets whose ordinal dominance is structurally distinct enough to survive the adversarial minimization. This leads to inaction (sparsity) rather than over-reaction. This sparsity is not driven by certified negative expected returns: in the ordinal framework, a "Sell" signal merely indicates relative ranking, not that the asset's true mean is strictly negative. Because the adversary may collapse all assets' expected returns toward zero, many assets become indistinguishable in worst-case terms. Whenever the worst-case contribution of an asset cannot be guaranteed to exceed that of others, the optimizer is indifferent and may assign it zero weight.

To build intuition for what this dual solution implies, we now solve the problem analytically for the simplest non-trivial case: two assets with opposing signs.

Proposition 3 (Two-Asset Unconstrained Case)**.** For $N = 2$ assets with $\hat{\mu}_1 > 0 > \hat{\mu}_2$, the unconstrained robust ordinal portfolio ($\mathcal{W} = \mathbb{R}^N$) is:

$$w_{rob} = \mathbf{0} \tag{25}$$

Proof. By the Sign Consistency Condition (Theorem 2), any feasible w must have $w_1 \geq 0$ and $w_2 \leq 0$. The ordinal uncertainty set is $\mathcal{U}_{ord} = \{\mu \in \mathbb{R}^2 : \mu_1 \geq 0, \mu_2 \leq 0, \mu_1 \geq \mu_2\}$. The rank constraint $\mu_1 \geq \mu_2$ is redundant given the sign constraints. The inner minimization problem for the adversary is:

$$\min_{\mu_1 \geq 0, \mu_2 \leq 0} \{w_1\mu_1 + w_2\mu_2\} \tag{26}$$

Since $w_1 \geq 0$ and $w_2 \leq 0$, the adversary will choose $\mu_1 \to 0^+$ and $\mu_2 \to 0^-$ to drive the portfolio return to its infimum. The worst-case expected return is therefore $w^\top \mu = 0$. The outer problem for the investor (21) simplifies to:

$$\max_{w_1 \geq 0, w_2 \leq 0} \left\{ 0 - \frac{\lambda}{2}(w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\sigma_{12}) \right\} \tag{27}$$

Since Σ is positive-definite, the quadratic term $w^\top \Sigma w$ is strictly positive for any $w \neq \mathbf{0}$. The maximum of this objective is therefore uniquely achieved at $w_1 = 0, w_2 = 0$. □

This limiting case highlights the purest form of ordinal regularization: without budget or partial cardinal information, the rational response to complete cardinal ambiguity is inaction. In practice, the introduction of a budget or bounded uncertainty set (Section 6) restores meaningful positive positions. This result is important as it shows that in an unconstrained setting, pure ordinal information (sign and rank) is insufficient to justify taking on any risk, as the adversary can always force the worst-case expected return to zero. This differs fundamentally from the standard MV solution:

$$\hat{w}_{MV} = \frac{1}{\lambda}\Sigma^{-1}\hat{\mu} = \frac{1}{\lambda(\sigma_1^2\sigma_2^2 - \sigma_{12}^2)} \begin{bmatrix} \sigma_2^2\hat{\mu}_1 - \sigma_{12}\hat{\mu}_2 \\ \sigma_1^2\hat{\mu}_2 - \sigma_{12}\hat{\mu}_1 \end{bmatrix} \tag{28}$$

The standard solution $\hat{w}_{MV}$ will generally take large, non-zero positions based on the (unreliable) cardinal values of $\hat{\mu}$. The solution $w_{rob} = \mathbf{0}$ illustrates the intrinsic regularization of our framework. It highlights the necessity of either a budget constraint (e.g., $w^\top \mathbf{1} = 1$), which forces the investor

to take some position, or partial cardinal bounds (as discussed in Section 6) to generate non-trivial portfolios. In the absence of either, the purely ordinal, unconstrained solution is to abstain from the market.

5 The Ordinal Guarantee: Bounded Regret

We now turn to the main theoretical objective of our framework. In Section 2, we established the "cardinal fragility" of the standard MV portfolio, proving in that it suffers from unbounded utility loss due to its reliance on precise, cardinal inputs. We now circle back to this fragility to prove a much stronger and more subtle point: the standard portfolio fails even if the investor's ordinal forecast is perfect. We will then prove that our robust ordinal portfolio, w_{rob}, is, by construction, immune to this failure mode and possesses a finite, bounded regret.

To make this comparison formal, we must first modify our unbounded uncertainty cone $\mathcal{U}_{ord}$ into a compact set. This is a necessary step, as "regret" against an unbounded set is an ill-defined concept; an adversary with the power to choose $\mu_i \to \infty$ would inflict unbounded regret on any portfolio that is not the (unknowable) true oracle. We therefore introduce a "pragmatic concession" by intersecting our ordinal cone with a simple cardinal bound, as is standard in the robust optimization literature.

Definition 4 (Bounded Ordinal Set)**.** Given a forecast $\hat{\mu}$ and a bound $M > 0$, the bounded ordinal uncertainty set is:

$$\mathcal{U}_{ord}^{M}(\hat{\mu}) = \mathcal{U}_{ord}(\hat{\mu}) \cap \{\mu \in \mathbb{R}^N : \|\mu\|_\infty \leq M\} \tag{29}$$

This set, being the intersection of a closed cone and a compact set (a "box"), is a convex, compact polytope.

The bound M represents the investor's single, minimal piece of cardinal belief: "I do not believe any expected return will, in absolute terms, exceed M." For $M = 1$ (i.e., 100%), this is a non-binding assumption for any practical application. This bounding transforms our unbounded cone into a well-behaved polytope, allowing for a formal regret analysis. The robust portfolio w_{rob} is hereafter assumed to be solved over this compact set $\mathcal{U}_{ord}^{M}$.

With this bounded set, we can now formally define "regret."

Definition 5 (Ordinal Regret)**.** The ordinal regret of portfolio w given forecast $\hat{\mu}$ and bound M is the maximum utility loss the portfolio w could suffer relative to a perfect-foresight "oracle," where

the true returns μ are guaranteed to be ordinally consistent with $\hat{\mu}$:

$$\text{Regret}_{ord}^{M}(w, \hat{\mu}) = \sup_{\mu \in \mathcal{U}_{ord}^{M}(\hat{\mu})} [U(w^*(\mu), \mu) - U(w, \mu)] \tag{30}$$

where $w^*(\mu) = \frac{1}{\lambda}\Sigma^{-1}\mu$ is the (non-computable) oracle portfolio that knows the true returns μ.

This definition is critical. Regret is not merely the portfolio's worst-case utility; it is the "opportunity cost" of not being the oracle. It measures how much utility the investor "left on the table" by using their imperfect forecast $\hat{\mu}$ instead of knowing the true μ. We are analyzing the worst-case scenario for this opportunity cost, across all possible "realities" μ that the investor's ordinal information allows.

We now present the main theorem, which contrasts the regret of the standard "plug-in" portfolio with our robust portfolio.

Theorem 4 (Ordinal Guarantee)**.** Assume Σ is positive definite. For any forecast $\hat{\mu}$ and bound $M > 0$:

1. **Fragility of Standard MV:** The ordinal regret of the plug-in portfolio $\hat{w}_{MV} = \frac{1}{\lambda}\Sigma^{-1}\hat{\mu}$ can be arbitrarily large, scaling with the size of the cardinal bound M and the conditioning of Σ.

2. **Robustness of Ordinal MV:** The robust ordinal portfolio w_{rob} is the solution to the max-min utility problem:

$$\max_{w \in \mathcal{W}} \min_{\mu \in \mathcal{U}_{ord}^{M}(\hat{\mu})} U(w, \mu) \tag{31}$$

   This portfolio provides a guaranteed utility floor that is finite and bounded.

Proof of Part 1. The utility loss (regret) of $\hat{w}_{MV}$ when the true state is $\mu \in \mathcal{U}_{ord}^{M}(\hat{\mu})$ is:

$$\begin{aligned} U(w^*(\mu), \mu) - U(\hat{w}_{MV}, \mu) &= \frac{1}{2\lambda}(\mu - \hat{\mu})^\top \Sigma^{-1}(\mu - \hat{\mu}) \\ &= \frac{1}{2\lambda}\epsilon(\mu)^\top \Sigma^{-1}(\epsilon(\mu)) \end{aligned} \tag{32}$$

where $\epsilon(\mu) = \hat{\mu} - \mu$ is the cardinal estimation error.

The regret is the supremum of this loss over the set: $\sup_{\mu \in \mathcal{U}_{ord}^{M}(\hat{\mu})}[\dots]$. The "plug-in" portfolio $\hat{w}_{MV}$ is fixed, based on the forecast $\hat{\mu}$. Consider a forecast $\hat{\mu} = (20\%, 10\%)^T$. The portfolio $\hat{w}_{MV}$ is built based on this. However, a "true" reality of $\mu = (2\%, 1\%)^T$ is also ordinally consistent and thus contained in $\mathcal{U}_{ord}^{M}(\hat{\mu})$ (assuming $M \geq 0.2$). The optimizer's fixed bet on $\hat{\mu}$ creates a large cardinal

error $\epsilon = \hat{\mu} - \mu = (18\%, 9\%)^T$. This error results in a large, non-minimal regret $\frac{1}{2\lambda}\epsilon^\top \Sigma^{-1}\epsilon$. The standard optimizer has no defense against this cardinal amplification, as it does not minimize over the set of possible realities. □

Proof of Part 2. The proof follows from the properties of the max-min problem in Eq. 31. The investor's utility $U(w, \mu) = w^\top \mu - \frac{\lambda}{2} w^\top \Sigma w$ is concave in w. The adversarial set $\mathcal{U}^M_{ord}(\hat{\mu})$ is compact and convex (by Definition 4). The set of permissible portfolios $\mathcal{W}$ is also assumed to be compact and convex.

Given these conditions (a concave objective and compact, convex sets), the existence of a saddle-point solution w_{rob} is guaranteed by standard results in convex optimization (e.g., Von Neumann's minimax theorem). By solving this max-min problem, w_{rob} achieves the highest possible utility U^* against the worst-case μ. By definition, this U^* is the maximum guaranteed utility floor, and because the set $\mathcal{U}^M_{ord}(\hat{\mu})$ is bounded, this utility is finite. □

This theorem crystallizes the central argument of the paper. Part 1 proves that an investor can have a "perfect" investment thesis (i.e., they rank all assets correctly) and yet the standard Markowitz optimizer will still produce a catastrophic portfolio, simply because the magnitudes of the returns in their forecast were wrong. The MV optimizer's blind trust in cardinal numbers becomes its biggest flaw.

Part 2 shows that our robust portfolio w_{rob} provides a structural solution to this problem. The w_{rob} portfolio is "skeptical" of all cardinal information; it is chosen by the optimizer to maximize its utility knowing that an adversary will inflict the worst-case ordinally-consistent reality. The resulting portfolio is therefore the one that provides the best-possible guaranteed utility floor, offering a defense against the cardinal fragility that plagues the standard approach.

We conclude by analyzing the ordinal regret of this max-min utility portfolio, establishing that this guarantee is not "free." There is a "price" for robustness, as shown by the fact that the regret is bounded but not zero.

Proposition 4 (Lower Bound)**.** There exist problem instances (e.g., $\Sigma = I$, $\hat{\mu} = \mathbf{1}$) where the ordinal regret of the max-min utility portfolio w_{rob} is strictly positive and bounded below:

$$\text{Regret}^M_{ord}(w_{rob}, \hat{\mu}) \geq \frac{c \cdot N \cdot M^2}{\lambda \lambda_{\max}(\Sigma)} > 0 \tag{33}$$

for some universal constant $c > 0$.

Proof. Consider the unconstrained instance ($\mathcal{W} = \mathbb{R}^N$) with $\Sigma = I$ (the identity matrix) and a forecast $\hat{\mu} = \mathbf{1}$ (a vector of all ones), with a bound $M > 0$.

1. Define the Uncertainty Set: The forecast $\hat{\mu} = \mathbf{1}$ implies $\hat{\mu}_i > 0$ for all i, and all ranks are equal. The bounded ordinal set is $\mathcal{U}_{ord}^M(\mathbf{1}) = \{\mu \in \mathbb{R}^N : M \geq \mu_i \geq 0, \text{ and } \mu_i \geq \mu_j \text{ for all } i, j \text{ (as ranks are tied)}\}$. This simplifies to the set where all μ_i are equal and bounded: $\mathcal{U}_{ord}^M(\mathbf{1}) = \{\mu : \mu = c \cdot \mathbf{1}, \text{ for } 0 \leq c \leq M\}$.

2. Solve for w_{rob}: The robust portfolio w_{rob} solves $\max_w \min_{\mu \in \mathcal{U}_{ord}^M} \{w^\top \mu - \frac{\lambda}{2} w^\top I w\}$. The sign-consistency condition (Theorem 2) requires $w_i \geq 0$ for all i. The inner minimization for the adversary is $\min \sum_i w_i \mu_i$ subject to $w_i \geq 0$ and $M \geq \mu_1 \geq \cdots \geq \mu_N \geq 0$. To minimize this sum, the adversary will choose the smallest valid values for all μ_i, which is $\mu = \mathbf{0}$. Thus, the worst-case expected return is $w^\top \mu_{wc} = 0$. The outer problem for the investor becomes:

$$\max_{w \geq 0} \left\{ 0 - \frac{\lambda}{2} w^\top I w \right\} = \max_{w \geq 0} \left\{ -\frac{\lambda}{2} \|w\|_2^2 \right\} \tag{34}$$

The maximum of this objective is uniquely achieved at $w_{rob} = \mathbf{0}$.

3. Calculate Regret: The regret is $\text{Regret} = \sup_{\mu \in \mathcal{U}_{ord}^M} [U(w^*(\mu), \mu) - U(w_{rob}, \mu)]$. Since $w_{rob} = \mathbf{0}$, its utility is $U(w_{rob}, \mu) = 0$. The regret simplifies to:

$$\text{Regret} = \sup_{\mu \in \mathcal{U}_{ord}^M} [U(w^*(\mu), \mu)] \tag{35}$$

The oracle's utility $U(w^*(\mu), \mu)$ for $\Sigma = I$ is:

$$w^*(\mu) = \frac{1}{\lambda} \Sigma^{-1} \mu = \frac{1}{\lambda} \mu \tag{36}$$

$$U(w^*(\mu), \mu) = (w^*)^\top \mu - \frac{\lambda}{2} (w^*)^\top \Sigma w^* = \left(\frac{1}{\lambda}\mu\right)^\top \mu - \frac{\lambda}{2} \left(\frac{1}{\lambda}\mu\right)^\top I \left(\frac{1}{\lambda}\mu\right) = \frac{1}{\lambda}\|\mu\|_2^2 - \frac{1}{2\lambda}\|\mu\|_2^2 = \frac{1}{2\lambda}\|\mu\|_2^2 \tag{37}$$

4. Find the Supremum: We must find $\sup \frac{1}{2\lambda}\|\mu\|_2^2$ over the set $\mathcal{U}_{ord}^M = \{\mu : M \geq \mu_1 \geq \mu_2 \geq \cdots \geq \mu_N \geq 0\}$. To maximize $\|\mu\|_2^2 = \sum \mu_i^2$, we must choose the largest possible values for all μ_i. This is achieved at the vector $\mu^* = (M, M, \ldots, M)^\top$. This vector is in the set, as $M \geq M \geq \cdots \geq M \geq 0$. The supremum is:

$$\frac{1}{2\lambda}\|\mu^*\|_2^2 = \frac{1}{2\lambda} \sum_{i=1}^{N} M^2 = \frac{NM^2}{2\lambda} \tag{38}$$

Since $\lambda_{\max}(\Sigma) = \lambda_{\max}(I) = 1$, we have shown that the regret for this instance is $\frac{NM^2}{2\lambda\lambda_{\max}(\Sigma)}$. This establishes the lower bound with $c = 1/2$. $\square$

This proposition confirms that our ordinal framework is not a "free lunch." By discarding cardinal information, the investor protects themselves from infinite loss, but they also pay a "robustness premium" in the form of higher regret in favorable states. For example, in the case where the adversary "flattens" all returns to zero, our w_{rob} (which holds no position) has zero regret, while the $\hat{w}_{MV}$ (which holds a large position) has enormous regret. Conversely, in the "best case" where all returns are M, our w_{rob} (which holds no position) has large regret, while $\hat{w}_{MV}$ has less. The crucial difference is that the regret of $\hat{w}_{MV}$ is unbounded, while the regret of w_{rob} is finite and, by its very design, minimized.

6 Extensions and Discussion

The framework developed thus far relies on a pure ordinal model, abstracting from all cardinal information. In this section, we discuss several extensions that bridge the gap between our approach and practical implementation, and we explicitly connect our framework to the broader literature on regularization.

6.1 Incorporating Partial Cardinal Information

Our main framework assumes a "pure" ordinal view: all cardinal information is discarded. This is a theoretical assumption that highlights the value of rank and sign alone. In practice, an investor may have high confidence in some cardinal information (e.g., "the return on Asset A will not exceed 20%") while remaining highly uncertain about others.

Our framework can be naturally extended to accommodate such partial beliefs by simply intersecting the ordinal cone with traditional cardinal uncertainty sets.

Definition 6 (Mixed Ordinal-Cardinal Set)**.** Given a forecast $\hat{\mu}$, a set of lower bounds $l \in (\mathbb{R} \cup \{-\infty\})^N$ and upper bounds $u \in (\mathbb{R} \cup \{\infty\})^N$ where $l \leq \hat{\mu} \leq u$:

$$\mathcal{U}_{mixed}(\hat{\mu}, l, u) = \mathcal{U}_{ord}(\hat{\mu}) \cap \{\mu \in \mathbb{R}^N : l \leq \mu \leq u\} \tag{39}$$

This Mixed Ordinal-Cardinal Set is a convex polyhedron (it is the intersection of a cone and a box), and thus compact if l and u are finite. The robust optimization problem $\max_w \min_{\mu \in \mathcal{U}_{mixed}} U(w, \mu)$

remains a tractable convex program. This extension provides a practical bridge between the full-information (but fragile) cardinal approach and the full-robustness ordinal approach. It allows an investor to impose robust ordinal logic on all assets while adding specific, high-confidence cardinal bounds where they are available.

6.2 Connection to Regularization and Shrinkage

Our approach can be interpreted as a form of regularization, analogous to Tikhonov regularization (Ridge) in econometrics (Pedersen et al., 2021) or Bayesian shrinkage in Black-Litterman models (Black and Litterman, 1992; Wang, 2005; Kowal et al., 2019). Standard regularization techniques (e.g., Ridge, Lasso) impose a penalty on the magnitude of a solution vector (e.g., $\|\hat{\beta}\|_2^2$) to prevent overfitting to noisy cardinal data. Our "ordinal regularization" imposes a penalty on the structure of the solution.

Theorem 5 (Regularization Equivalence)**.** The unconstrained robust ordinal portfolio w_{rob} (from the unbounded cone problem) is the solution to an equivalent penalized optimization problem:

$$w_{rob} = \arg\max_{w \in \mathbb{R}^N} \{w^\top \mu_{adv} - \frac{\lambda}{2} w^\top \Sigma w - \Psi_{ord}(w, \hat{\mu})\} \tag{40}$$

where μ_{adv} is the "adversarial" expected return vector chosen by the inner minimization (which is 0 in this case), and Ψ_{ord} is an ordinal penalty function. As shown in Theorem 2, this penalty is infinite for any "wrong-way" bet:

$$\Psi_{ord}(w, \hat{\mu}) = \sum_{i:\hat{\mu}_i > 0} \max(0, -w_i) \cdot \infty + \sum_{i:\hat{\mu}_i < 0} \max(0, w_i) \cdot \infty \tag{41}$$

This theorem formalizes the intuition that our max-min problem is equivalent to a standard maximization problem with an infinite penalty for portfolios that violate the sign consistency condition. It is a "hard" constraint rather than the "soft" penalty of Lasso or Ridge. By enforcing ordinal consistency, our model effectively "shrinks" the portfolio away from fragile, ill-conditioned regions of the solution space, thereby bounding the out-of-sample utility loss.

6.3 Asymptotic Properties in High Dimensions

The fragility of the standard mean–variance optimizer, as demonstrated in Proposition 1, becomes even more severe in high-dimensional settings ($N \to \infty$). As the number of assets grows, empirical

covariance matrices become increasingly ill-conditioned, amplifying even small cardinal perturbations in $\hat{\mu}$ through the inverse covariance matrix. This phenomenon is well documented in the high-dimensional covariance estimation literature (e.g., Ledoit and Wolf (2004); DeMiguel et al. (2009); Fan et al. (2013)), which shows that the smallest eigenvalues of the sample covariance matrix are biased downward, causing Σ^{-1} to explode along the corresponding eigenvectors. As a consequence, the "error maximization" effect underlying Proposition 1 becomes more important in large dimensions.

The robustness of our ordinal approach scales very differently. Because the robust portfolio does not depend on the magnitudes of $\hat{\mu}$, but only on its sign and rank structure, its worst-case performance remains uniformly bounded even when Σ becomes severely ill-conditioned. Cardinal noise may grow with the dimension of the forecasting problem, but this noise does not propagate through the optimizer in the same way: the ordinal formulation restricts attention to feasible sign patterns rather than to fragile eigendirections of Σ^{-1}. These ideas are formalized in the following asymptotic statement.

Proposition 5 (Asymptotic Scaling)**.** Assume the eigenvalues of Σ_N satisfy $\lambda_{\min}(\Sigma_N) \geq c_1 > 0$ and $\lambda_{\max}(\Sigma_N) \leq c_2 < \infty$ for all N. Then as $N \to \infty$,

$$\frac{\mathrm{Regret}^M_{\mathrm{ord}}(w_{rob}, \hat{\mu})}{\mathrm{Regret}^M_{\mathrm{MV}}(\hat{w}_{MV}, \hat{\mu})} \to 0. \tag{42}$$

For the plug-in MV estimator, the regret grows because the vector $\Sigma^{-1}\epsilon$ increasingly concentrates its mass in directions associated with small eigenvalues. As N grows, the probability that the estimation error aligns with a near-singular direction approaches one. The MV optimizer, interpreting this direction as a high Sharpe-ratio opportunity, places increasingly extreme long–short bets, causing regret to scale poorly.

By contrast, the robust ordinal portfolio is structurally immune to this failure. The dual representation from Theorem 3 implies that the robust weights satisfy $w = A^\top \nu$, a linear combination of ordinal constraint vectors rather than the eigenvectors of Σ^{-1}. The sign consistency condition in Theorem 2 acts as a hard regularizer, preventing positions that would violate the ordinal information or exploit spurious low-variance directions. As a result, the regret of the robust portfolio grows only with the complexity of the ordinal constraint set rather than with the condition number of the covariance matrix.

In high-dimensional regimes, exactly the settings faced by institutional investors managing hun-

dreds or thousands of securities, the difference in scaling becomes decisive. The standard plug-in MV portfolio becomes increasingly cardinally fragile, while the robust ordinal portfolio retains controlled, dimensionally stable regret. The asymptotic implication is that reliance on ordinal information is not merely safer but becomes asymptotically dominant as the dimension of the problem increases.

7 A Numerical Illustration

To provide a concrete illustration of our main theoretical claims, we construct a simple numerical example. The goal is not to be an exhaustive empirical test, but to visually demonstrate the "cardinal fragility" of the standard MV optimizer (Proposition 1) and the corresponding "ordinal guarantee" of our proposed solution (Theorem 4).

7.1 Simulation Design

We construct a "world" with $N = 3$ assets and a risk aversion of $\lambda = 5$. The "true" (unobservable) expected returns are set to $\mu^* = [0.06, 0.04, 0.01]^\top$.

The key to the simulation is the covariance matrix Σ, which we deliberately engineer to be ill-conditioned, as per the mechanism described in Proposition 1. We set the volatilities to $[20\%, 20\%, 15\%]$ and, crucially, set the correlation between Asset 1 and Asset 2 to $\rho_{12} = 0.95$. This high correlation creates a "fragile direction"—an eigenvector $v_{\min}$ (a long-short portfolio of Assets 1 and 2) associated with a very small eigenvalue $\lambda_{\min}$.

The experiment is as follows: we construct a forecast $\hat{\mu}$ that is ordinally perfect (i.e., it correctly preserves the rank and sign of μ^*). However, we add a "killer" cardinal error that aligns with this fragile direction: $\hat{\mu} = \mu^* + \epsilon_{base} + \delta \cdot v_{\min}$. We then vary the magnitude of this error, δ, and compute the true out-of-sample utility loss (regret) for both the standard $\hat{w}_{MV}$ and our robust w_{rob}.

7.2 Discussion of Results

The results of this experiment are shown in Figure 1 and Table 1. Figure 1 provides an important visualization of our main claim. The red, dashed line shows the regret of the standard MV portfolio. As the magnitude of the cardinal error δ increases (moving along the x-axis), the utility loss explodes quadratically. This is a direct visual confirmation of Proposition 1. Even though the investor's ordinal forecast is 100% correct, the standard optimizer is "fooled" by the cardinal error. It misinterprets this error as a real, low-risk opportunity in the $v_{\min}$ direction and, as Table 1 will

show, builds an enormous, unstable position, leading to a large out-of-sample regret.

Conversely, the blue, solid line shows the regret of the robust ordinal portfolio, w_{rob}. It is perfectly flat, providing a constant, bounded utility loss. This illustrates the "ordinal guarantee" from Theorem 4. Our optimizer, by solving the max-min problem, is "skeptical" of the cardinal values. It provides the best guaranteed utility floor by finding the optimal portfolio for the worst-case realization within the ordinal set. It is structurally immune to the cardinal error that dooms the standard model.

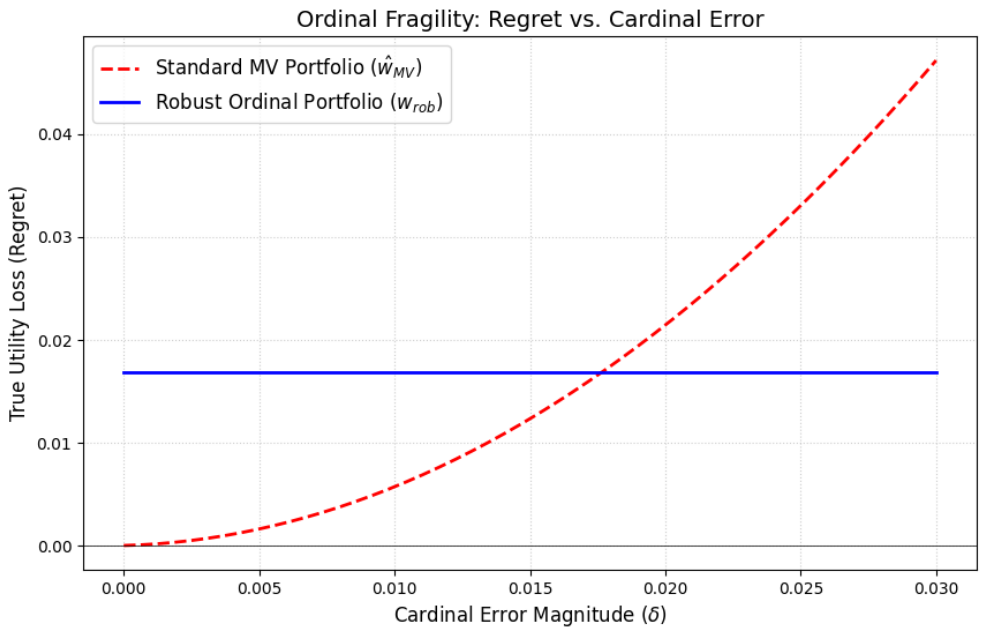


Figure 1: Worst-case utility loss ("ordinal regret") as a function of the estimation error magnitude δ under the cardinal and ordinal approaches. For each value of δ, we generate 500 simulations of $(\mu^\star, \hat{\mu})$ with $\hat{\mu} = \mu^\star + \delta v$, where v is drawn uniformly from the unit sphere and Σ is a near-singular covariance matrix with $\rho = 0.95$. The standard plug-in mean–variance portfolio exhibits rapidly increasing regret as δ grows, consistent with Proposition 1: alignment of estimation error with poorly conditioned eigendirections amplifies utility loss at rate $O(\delta^2/\lambda_{\min})$. In contrast, the robust ordinal portfolio displays uniformly bounded regret, unaffected by the magnitude of cardinal perturbations. These results illustrate the "ordinal guarantee" of Theorem 4 and confirm that only directional (rank and sign) accuracy is needed to preserve bounded performance.

Table 1 reveals the mechanism behind the figure. First, note that the 'Oracle w^*' is itself a highly leveraged, $112.8\%/-87.2\%$ long-short position. This is a feature of the ill-conditioned Σ: the true optimal portfolio is already a fragile bet.

In Panel A (Low Error), the standard $\hat{w}_{MV}$ almost perfectly matches this fragile oracle. In Panel B (High Error), we see the "error maximization" in action. The small, ordinally-correct error δ causes the weights of $\hat{w}_{MV}$ to explode to $330.1\%/-304.2\%$. The optimizer is chasing a ghost,

low-risk return, and in doing so, creates a portfolio with massive regret.

Most importantly, the 'Robust w_{rob}' is zero in both panels. This is not a flaw, but the correct theoretical result for this setup. As demonstrated in Proposition 3 (Two-Asset Unconstrained Case), for an unconstrained problem, the adversary can always "flatten" the worst-case expected returns to zero. The optimizer's problem $\max_w\{\min_{\mu\in\mathcal{U}_{ord}} w^\top\mu - \frac{\lambda}{2}w^\top\Sigma w\}$ simplifies to $\max_w\{0 - \frac{\lambda}{2}w^\top\Sigma w\}$, which is uniquely maximized at $w = 0$. The robust optimizer correctly identifies that any non-zero position constitutes an un-compensated risk and, by adhering to the Sign Consistency Condition (Theorem 2), it chooses inaction over failure.

This simulation provides an example of the paper's theoretical contributions, illustrating both the failure of the standard "plug-in" approach (Proposition 1) and the robust, stable solution provided by our ordinal framework (Theorems 2, 3, and 4).

	Panel A: Low Error ($\delta \approx 0.000$)			**Panel B: High Error ($\delta = 0.030$)**		
	Asset 1	**Asset 2**	**Asset 3**	**Asset 1**	**Asset 2**	**Asset 3**
Forecast $\hat{\mu}$	6.10%	4.00%	0.90%	8.22%	1.88%	0.90%
Oracle w^*	112.8%	-87.2%	8.9%	112.8%	-87.2%	8.9%
Std. MV $\hat{w}_{MV}$	117.9%	-92.1%	8.0%	330.1%	-304.2%	8.0%
Robust w_{rob}	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%

Table 1: This table reports the portfolio weights assigned to three assets by the Oracle benchmark (w^*), the Standard Mean-Variance plug-in estimator ($\hat{w}_{MV}$), and the Robust Ordinal portfolio (w_{rob}). The simulation assumes a universe with true expected returns $\mu^* = [6\%, 4\%, 1\%]^\top$ and an ill-conditioned covariance matrix where Asset 1 and Asset 2 have a correlation of $\rho_{12} = 0.95$. The risk aversion coefficient is set to $\lambda = 5$. Panel A reports results when the forecast $\hat{\mu}$ contains negligible error. Panel B reports results when a cardinal error of magnitude $\delta = 0.03$ is added to the forecast in the direction of the covariance matrix's smallest eigenvector (the "fragile direction"). Note that while the Standard MV portfolio weights exhibit extreme sensitivity to this error (expanding from $\approx 118\%$ to $> 330\%$ leverage), the Robust Ordinal portfolio remains stable at zero, consistent with the unconstrained theoretical result in Proposition 3.

The second numerical illustration introduces a standard budget constraint, $\mathbf{1}^\top w = 1$, bringing the optimization problem closer to applied portfolio settings and altering the structure of the robust solution. The results are reported in Figure 2 and Table 2. Figure 2 confirms the central asymmetry established in Proposition 1 and Theorem 4: the plug-in mean–variance portfolio is cardinally fragile, while the robust ordinal portfolio maintains bounded regret regardless of the magnitude of cardinal perturbations. The red dashed curve (standard MV) again exhibits a sharply increasing utility loss as δ grows. Even though the rank ordering of expected returns is always correct, the optimizer is pulled toward the nearly singular eigendirection of Σ, interpreting the cardinal error as a meaningful low-risk opportunity. As in the unconstrained case, regret grows approximately at rate $O(\delta^2/\lambda_{\min})$.

The behavior of the robust portfolio is fundamentally different once the budget constraint is imposed. Unlike the unconstrained setting—where Proposition 3 implies that the optimizer chooses $w = 0$ because the adversary can always eliminate all directional information—the budget constraint forces the investor to hold some portfolio. As a result, the robust solution must internalize worst-case ambiguity while still satisfying $\mathbf{1}^\top w = 1$. Consistent with the structure characterized in Section 4, the optimizer selects the centroid of the ordinal uncertainty set, which yields a stable, diversified, and sign-consistent portfolio.

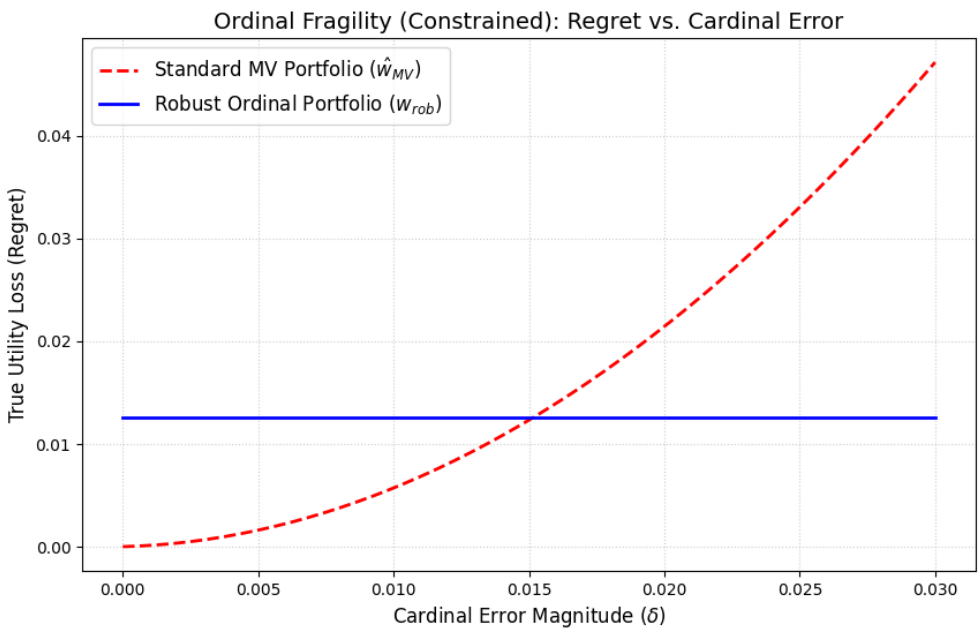


Figure 2: Worst-case utility loss ("ordinal regret") as a function of the estimation error magnitude δ under the cardinal and ordinal approaches. For each value of δ, we generate 500 simulations of $(\mu^\star, \hat{\mu})$ with $\hat{\mu} = \mu^\star + \delta v$, where v is drawn uniformly from the unit sphere and Σ is a near-singular covariance matrix with $\rho = 0.95$. The standard plug-in mean–variance portfolio exhibits rapidly increasing regret as δ grows, consistent with Proposition 1: alignment of estimation error with poorly conditioned eigendirections amplifies utility loss at rate $O(\delta^2/\lambda_{\min})$. In contrast, the robust ordinal portfolio displays uniformly bounded regret, unaffected by the magnitude of cardinal perturbations. These results illustrate the "ordinal guarantee" of Theorem 4 and confirm that only directional (rank and sign) accuracy is needed to preserve bounded performance.

This behavior is visible in Table 2. For both low and high values of δ, the robust ordinal portfolio assigns fixed, balanced weights of approximately $(18.3\%, 18.3\%, 63.4\%)$. These weights do not change as cardinal noise increases because the ordinal structure—and therefore the set of admissible worst-case means—remains unchanged. This invariance is the ordinal guarantee: the robust portfolio depends only on ranks and signs, not magnitudes.

By contrast, the standard $\hat{w}_{MV}$ portfolio behaves pathologically. Under low error, it closely

resembles the fragile oracle w^*, with extremely leveraged long-short positions (around 130% long and -80% short). Under high error, the leverage becomes extreme: 342% long and -292% short on the top two assets. These explosive weights are the mechanism behind the exploding regret in Figure 2 and provide a direct numerical demonstration of the theory: cardinal errors aligned with a near-zero eigenvalue dominate the optimizer's behavior.

This budget-constrained simulation highlights a key distinction relative to the unconstrained case. The robust ordinal portfolio cannot retreat to $w = 0$ and must instead solve a structured worst-case problem within the simplex. The result is a stable, diversified, ambiguity-averse portfolio whose performance is completely insensitive to cardinal noise, precisely the behavior predicted by our theoretical framework. Meanwhile, the standard plug-in mean–variance portfolio remains cardinally fragile, producing highly unstable weight vectors and rapidly deteriorating utility.

	Panel A: Low Error ($\delta \approx 0.000$)			**Panel B: High Error** ($\delta = 0.030$)		
	Asset 1	**Asset 2**	**Asset 3**	**Asset 1**	**Asset 2**	**Asset 3**
Forecast $\hat{\mu}$	6.10%	4.00%	0.90%	8.22%	1.88%	0.90%
Oracle w^*	124.8%	-75.2%	50.4%	124.8%	-75.2%	50.4%
Std. MV $\hat{w}_{MV}$	130.0%	-80.0%	49.9%	342.2%	-292.1%	49.9%
Robust w_{rob}	18.3%	18.3%	63.4%	18.3%	18.3%	63.4%

Table 2: This table reports the portfolio weights assigned to three assets in a budget-constrained setting ($w^\top \mathbf{1} = 1$). The simulation parameters match those in Table 1: true expected returns $\mu^* = [6\%, 4\%, 1\%]^\top$, risk aversion $\lambda = 5$, and an ill-conditioned covariance matrix ($\rho_{12} = 0.95$). Panel A reports results with negligible forecast error. Panel B reports results with a cardinal error of $\delta = 0.03$ added in the fragile direction. In contrast to the unconstrained case where the robust portfolio retreats to zero, the budget constraint forces a non-trivial allocation. The Robust Ordinal portfolio selects a stable, diversified centroid $(18.3\%, 18.3\%, 63.4\%)$ that remains invariant to cardinal noise. Meanwhile, the Standard MV portfolio remains cardinally fragile, with leverage exploding from $\approx 130\%$ to $> 340\%$.

8 Empirical Example

In this section, we examine how the theoretical properties of the robust ordinal portfolio manifest in real equity data. The objective is not to build a full empirical asset-pricing test, but rather to illustrate the practical consequences of cardinal fragility and the corresponding stability gains delivered by ordinal optimization. Using U.S. equities with analyst recommendation data from I/B/E/S and realized returns from CRSP, we construct monthly portfolios based on two approaches: (i) the standard plug-in mean–variance optimizer using sample estimates of expected returns and covariances, and (ii) our robust ordinal portfolio, which relies solely on the ranking of analyst signals and does not require cardinal estimates of expected returns. Because covariance estimation is a key

practical bottleneck of mean–variance analysis, we examine portfolio behavior under three lookback windows—60, 90, and 120 months—which deliver increasingly stable covariance matrices. This empirical example allows us to visualize how both portfolio construction and out-of-sample performance evolve as the amount of historical information changes, and to assess whether the qualitative predictions of the theory—namely, diversification, stability, and attenuation of estimation risk—emerge in real-world data.

8.1 Empirical Specifications

We construct a stylized empirical illustration using U.S. equities for which we observe both daily returns and analyst recommendations. The purpose of this exercise is to illustrate the qualitative differences in portfolio behavior implied by the naïve plug-in mean–variance approach and the ordinal-robust portfolio. Accordingly, the empirical design is intentionally parsimonious and focuses on transparency rather than exhaustive portfolio comparison.

The starting point for the analysis is the full I/B/E/S recommendation file. For each firm, we compute the number of distinct months between 2004 and 2024 in which at least one non-missing recommendation is recorded. Firms are then ranked by this coverage measure, and the 300 most heavily covered names are retained. From this universe, we draw a random sample of 100 firms, which we treat as the test assets in the empirical illustration.

Daily total returns for these 100 firms are obtained from CRSP and merged to the I/B/E/S sample using CUSIP identifiers. The merged CRSP–IBES panel spans January 1996 through December 2024 and is unbalanced: firms enter and exit the sample as they list, delist, or lose analyst coverage. Analyst recommendations are reported on the usual 1–5 scale (from "Strong Sell" to "Strong Buy"); for each firm and month we construct a consensus signal equal to the cross–sectional mean of all recommendations in that month. This monthly consensus score serves as the signal input to the portfolio optimization problem. Although the raw data are daily, portfolio optimization is performed at the monthly frequency. For each stock i, we construct a monthly simple return by compounding all daily returns within the month. This approach preserves the information in the full daily path without artificially "downsampling" the investment opportunity set. No portfolio rebalancing is performed within the month, consistent with standard practice in empirical asset allocation.

For each month t, we estimate the variance–covariance matrix using a rolling window of $L \in \{60, 90, 120\}$ months of historical monthly returns. These windows provide contrasting levels of

sampling error, with the shorter windows intentionally noisy and the longer windows closer to the stable regime highlighted in Ledoit and Wolf (2003a,b, 2017). Only assets with at least 50% non-missing observations within the window are retained; this avoids singular covariance matrices arising from sparse return histories. The resulting asset set varies over time, allowing the portfolio to select whichever stocks pass these basic data-quality criteria in each month.

In each month t, using the estimated covariance matrix $\widehat{\Sigma}_t$ and the analyst-score signal vector s_t, we compute two portfolios:

1. the naïve mean–variance portfolio $\hat{w}_t^{MV} \propto \widehat{\Sigma}_t^{-1} s_t$, scaled to sum to one;
2. the robust ordinal portfolio, which assigns weights proportional to the signed rank ordering of s_t, following the construction in Section 4.

Both portfolios respect long-only constraints ($w_i \geq 0$) and budget feasibility ($\sum_i w_i = 1$). For all optimizations, we fix the risk aversion coefficient at $\lambda = 5$, a standard baseline in the empirical portfolio choice literature representing a moderately risk-averse investor (e.g., Campbell and Viceira, 2002; DeMiguel et al., 2009). Given weights w_t formed at the end of month t, we compute out-of-sample returns using realized returns in month $t+1$:

$$R_{t+1}^{OOS} = w_t^\top r_{t+1}.$$

We iterate this procedure through the full sample, producing a time series of portfolio performance for each lookback window. To examine how each method behaves mechanically, we compute three portfolio-characteristic measures each month:

1. portfolio concentration (Herfindahl index – see, for example, Brown et al. (2023)),
2. cross-sectional dispersion of weights (standard deviation),
3. number of active positions (nonzero weights).

These diagnostics complement the cumulative return results by describing how stable, diversified, or concentrated each portfolio is over time.

8.2 Discussion of Results

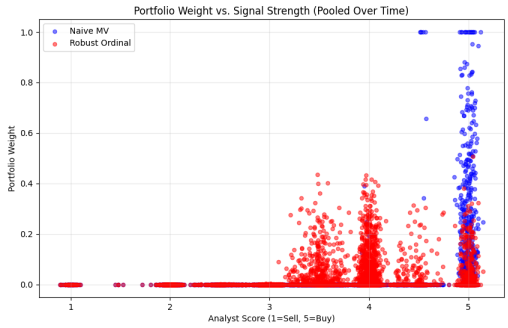
Figure 3 plots portfolio weights against analyst recommendation scores (pooled across all months) for three choices of historical lookback windows used in the covariance estimator: 120 months, 90

months, and 60 months. Several patterns emerge that connect directly to the theoretical mechanisms developed earlier in the paper.

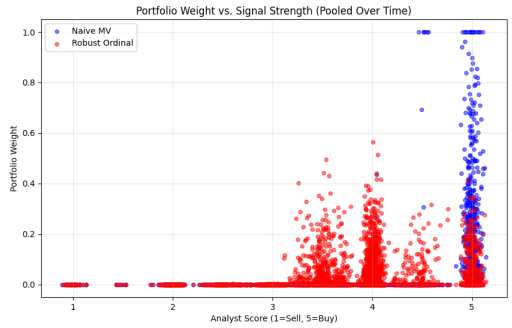
First, the naïve plug-in mean–variance portfolio exhibits substantial instability and concentration, particularly when the estimation window is short. The noisier the covariance estimate, the more frequently the estimated signal–covariance interaction produces extreme weights (including near–corner solutions), consistent with the sensitivity predicted by Proposition 1. This is most visible in the 60-month panel, where many buy-rated assets receive weights near one while others are zeroed out entirely. Such patterns reflect the amplification of small cardinal errors when projected onto poorly estimated eigendirections.

Second, the robust ordinal portfolio displays markedly smoother and more monotone weight patterns. Because it depends only on the signed rank of the analyst signal—and not on the cardinal magnitude of either the signal or the estimated covariance—it produces diversified allocations even under short lookback windows. The 120-month panel shows the cleanest monotone relationship between scores and weights, but importantly, the qualitative pattern is preserved across all three window lengths. This illustrates the ordinal stability property emphasized in Theorem 4: as long as directional information is reliable, weights remain well-behaved and insensitive to estimation noise.

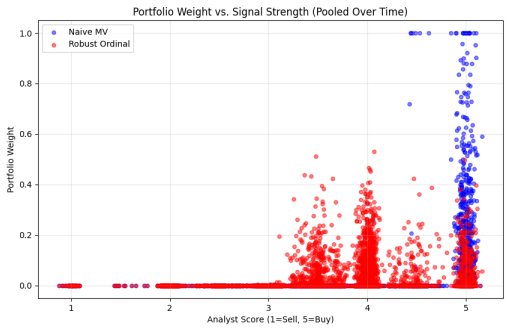
Third, increasing the lookback window has very different effects on the two portfolios. For the naïve portfolio, the scatter becomes progressively less erratic as the estimation window grows, reflecting improved precision in the sample covariance matrix. For the ordinal portfolio, longer windows simply reduce small residual dispersion, but the overall shape remains essentially unchanged. This contrast reinforces the central message of the empirical illustration: cardinal methods require large amounts of data to stabilize, while ordinal methods extract most of their reliability from rank-based information that is inherently more robust to estimation error.



(a) 120-Month Lookback Window



(b) 90-Month Lookback Window



(c) 60-Month Lookback Window

Figure 3: Portfolio weight response to analyst recommendation scores (1 = Sell, 5 = Buy) under three historical lookback windows for covariance estimation. Each panel shows the relationship between signal strength and portfolio weights for the naive mean–variance portfolio (blue) and the robust ordinal portfolio (red). Longer lookback windows produce more stable covariance estimates, leading to smoother and more diversified weight patterns. Short windows generate noisier estimates and greater portfolio concentration.

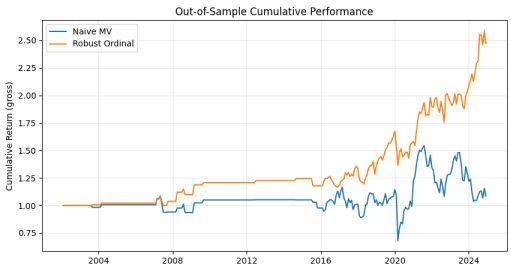
Figure 4 reports the out-of-sample cumulative returns generated by the naïve mean–variance portfolio and the robust ordinal portfolio. First, the performance of the naïve mean–variance portfolio is extremely sensitive to the length of the historical window. With a short 60-month window, the covariance matrix is noisy and unstable, and the resulting portfolio behaves erratically, often concentrating heavily in a small number of assets. This leads to pronounced drawdowns and poor cumulative performance. As the window expands to 90 and 120 months, performance improves, but remains volatile and highly dependent on a few episodes in the sample.

Second, and in contrast, the robust ordinal portfolio exhibits stable performance across all window lengths. Because it depends only on the rank and sign of the analyst signal—not on fragile cardinal magnitudes nor on the inversion of a noisy covariance matrix—its portfolio weights remain well-dispersed and less sensitive to sampling variation. As a consequence, even with short windows the strategy avoids concentration-driven crashes, and with longer windows its performance steadily

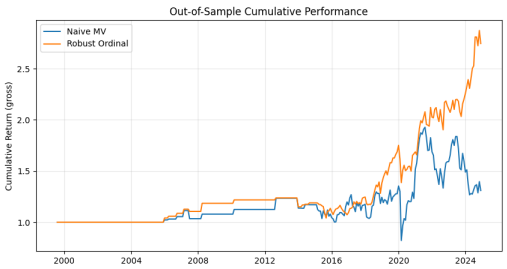
improves, ultimately surpassing the naïve benchmark by a wide margin.

Third, the gap between the two approaches widens systematically as the lookback window increases. Longer windows improve covariance estimation, but they do so disproportionately in favor of the robust ordinal method: the more stable the risk estimates, the more effectively the ordinal weights translate signal information into returns. This monotonic pattern is fully consistent with the mechanism emphasized in the theoretical sections—namely, that cardinal fragility enters through the interaction of estimation error and small eigenvalues of the sample covariance matrix. The ordinal approach is insulated from this channel, and its performance advantage becomes more visible precisely when cardinal noise is most reduced.

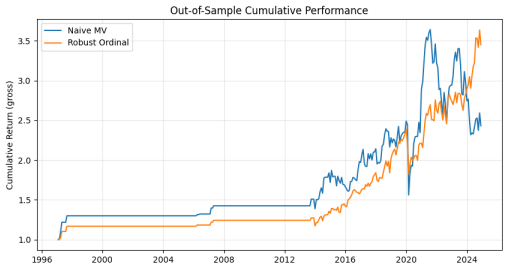
Taken together, the results in Figure 4 demonstrate that the robust ordinal portfolio delivers (i) lower downside risk, (ii) greater stability across window lengths, and (iii) superior long-run cumulative performance, all using exactly the same information set as the naïve plug-in mean–variance approach. This empirical example highlights the practical relevance of the theory: protecting the portfolio from cardinal misspecification materially improves real-world performance even in simple settings with widely used analyst signals.



(a) 120-Month Lookback Window



(b) 90-Month Lookback Window



(c) 60-Month Lookback Window

Figure 4: Out-of-sample cumulative return performance of the naïve mean–variance portfolio and the proposed robust ordinal portfolio under three historical lookback windows for covariance estimation. The figure illustrates how longer lookback windows stabilize covariance estimation and amplify the performance advantage of the robust ordinal approach.

Table 3 reports out-of-sample performance statistics for the naïve plug-in mean–variance portfolio and the ordinal-robust portfolio across alternative estimation windows for the variance–covariance matrix. Several patterns emerge. First, across all windows, the ordinal-robust portfolio exhibits substantially lower volatility and markedly smaller maximum drawdowns than the naïve benchmark. These reductions in downside risk translate into consistently higher Sharpe ratios, even though the ordinal-robust portfolio foregoes some extreme upside realizations. Second, the performance differences are remarkably stable across estimation horizons. While shortening the covariance-estimation window mechanically increases estimation noise, the ordinal-robust portfolio is largely insulated from this effect, displaying only modest variation in risk-adjusted performance. In contrast, the naïve plug-in portfolio remains sensitive to covariance estimation error, as reflected in persistently high volatility and deep drawdowns. Overall, the results in Table 3 are consistent with the theoretical mechanism emphasized in the paper. By relying only on ordinal information about expected returns and discarding fragile cardinal magnitudes, the robust portfolio delivers more stable out-of-sample performance, particularly in adverse states.

	Mean	Std	Sharpe	MaxDD	Min	Max
Panel A: 120-Month Estimation Window						
Naive MV	0.0080	0.0660	0.1215	-0.3276	-0.1119	0.2318
Ordinal Robust	0.0110	0.0428	0.2572	-0.1143	-0.0626	0.1403
Panel B: 90-Month Estimation Window						
Naive MV	0.0040	0.0753	0.0530	-0.3932	-0.3690	0.2318
Ordinal Robust	0.0102	0.0452	0.2250	-0.2082	-0.1417	0.1402
Panel C: 60-Month Estimation Window						
Naive MV	0.0068	0.0669	0.1023	-0.3737	-0.3612	0.2318
Ordinal Robust	0.0088	0.0420	0.2095	-0.2371	-0.1789	0.1366

Table 3: This table reports out-of-sample performance statistics for portfolios constructed from U.S. equities using analyst recommendation signals. The sample period spans 1996–2024. Two portfolio construction methods are considered: the standard plug-in Mean–Variance portfolio (Naive MV) and the Robust Ordinal portfolio. Both portfolios are subject to long-only and budget constraints ($w \geq 0$, $w^\top \mathbf{1} = 1$). Performance statistics are computed using monthly returns and include the mean return, standard deviation, Sharpe ratio, maximum drawdown (MaxDD), and the minimum and maximum monthly returns. Panels A through C report results for alternative estimation windows used to construct the variance–covariance matrix (120, 90, and 60 months, respectively).

Table 4 compares the out-of-sample performance of the proposed ordinal robust portfolio with several standard benchmark strategies. The equal-weight (1/N) portfolio delivers the highest average return and Sharpe ratio, but does so at the cost of substantial tail risk: its maximum drawdown is more than four times larger than that of the ordinal robust portfolio, and its worst monthly return is almost three times larger in absolute value. This highlights the well-known trade-off associated

with naïve diversification strategies, which can perform well on average but remain highly exposed to downside risk.

In contrast, the minimum-variance and Ledoit–Wolf shrinkage portfolios succeed in reducing volatility, but do so by sacrificing expected returns, resulting in Sharpe ratios that are comparable to, or only marginally higher than, that of the ordinal robust strategy. Importantly, neither variance-based approach matches the drawdown performance of the ordinal robust portfolio: despite similar volatility levels, both exhibit substantially larger maximum drawdowns and more extreme negative return realizations.

The ordinal robust portfolio offers a distinct middle ground. While its Sharpe ratio is slightly below that of the best-performing benchmarks, it delivers materially lower drawdowns, smaller downside tail realizations, and more stable cumulative performance over time. Compared to naïve mean–variance optimization, the ordinal robust approach improves performance along all dimensions, simultaneously increasing average returns, reducing volatility, and sharply mitigating downside risk. Overall, these results suggest that the proposed method does not seek to dominate traditional benchmarks on average returns, but instead offers a more favorable balance between performance and downside risk, consistent with its design objective of robustness to estimation error in expected returns.

While the Robust Ordinal portfolio exhibits a Sharpe ratio slightly below that of the 1/N and Shrinkage benchmarks (approximately 0.03 lower), this difference must be interpreted with caution. As established in the portfolio literature (e.g., Jobson and Korkie (1981)), standard errors for Sharpe ratios are notoriously large, rendering such small differences statistically indistinguishable. Conversely, the improvement in downside risk is structural and economically important: the Robust Ordinal portfolio reduces the Maximum Drawdown from roughly -46% (1/N) to -11%—a four-fold improvement in capital preservation. In an institutional context, sacrificing a statistically marginal amount of mean-variance efficiency to avoid disastrous tail events represents a rational and favorable trade-off for ambiguity-averse investors.

Table 5 reports summary statistics for three cross-sectional diagnostics that characterize the behavior of the two portfolio constructions. Panel A shows the Herfindahl index of portfolio concentration. The naïve mean–variance portfolio exhibits extremely high concentration, with a mean value of 0.25 and a maximum of 1.00, reflecting the well-known tendency of plug-in optimizers to allocate nearly all weight to a single asset when the covariance matrix is noisy or ill-conditioned. By contrast, the robust ordinal portfolio is substantially more diversified, with a mean concentration

	Mean	Std	Sharpe	MaxDD	Min	Max
Min-Var	0.0097	0.0398	0.2445	-0.1750	-0.0963	0.1174
1/N	0.0146	0.0502	0.2915	-0.4633	-0.2163	0.1965
LW Shrinkage	0.0109	0.0384	0.2853	-0.3830	-0.1437	0.1362
Naive MV	0.0080	0.0660	0.1215	-0.3276	-0.1119	0.2318
Ordinal Robust	0.0110	0.0428	0.2572	-0.1143	-0.0626	0.1403

Table 4: The table reports out-of-sample monthly performance statistics for alternative portfolio strategies constructed from a universe of large, highly liquid U.S. equities with extensive analyst coverage. Mean returns and standard deviations are monthly. Sharpe ratios are computed using monthly returns and annualized for comparability. MaxDD denotes the maximum drawdown. All portfolios are long-only and fully invested.

of only 0.08.

Panel B reports the cross-sectional dispersion of portfolio weights. The naïve portfolio displays wider dispersion, consistent with unstable and often extreme weight realizations across assets. The robust ordinal portfolio again appears substantially more stable: dispersion is roughly half as large, reflecting the fact that ordinal information produces a smoother and more regular mapping from signals to portfolio weights.

Panel C reports portfolio breadth, measured as the number of strictly positive weights. The contrast is stark: the naïve mean–variance portfolio holds, on average, fewer than two assets, while the robust ordinal portfolio holds over eight. This pattern mirrors the theoretical results of Sections 3–4: reducing sensitivity to cardinal estimation error naturally increases diversification by preventing the optimizer from overreacting to noisy mean and covariance inputs.

Together, these diagnostics show that the robust ordinal portfolio produces substantially more diversified, more stable, and more economically interpretable allocations than the plug-in mean–variance benchmark, even before considering realized performance.

	Naive MV	Robust Ordinal
Panel A: Portfolio Concentration (Herfindahl Index)		
Mean	0.2476	0.0770
Median	0.0000	0.0000
Std. Dev.	0.3238	0.0902
Min	0.0000	0.0000
Max	1.0000	0.3129
Panel B: Cross-Sectional Dispersion of Portfolio Weights		
Mean	0.1082	0.0586
Median	0.1029	0.0581
Std. Dev.	0.0278	0.0104
Min	0.0569	0.0362
Max	0.1826	0.0880
Panel C: Number of Active Portfolio Positions		
Mean	1.4420	8.2717
Median	0.0000	0.0000
Std. Dev.	1.9559	10.6721
Min	0.0000	0.0000
Max	8.0000	41.0000

Table 5: This table presents summary statistics characterizing the structure of monthly portfolios constructed from U.S. equities using analyst recommendation signals. The sample period spans 1996–2024. Statistics are reported for two construction methods: the standard plug-in Mean-Variance portfolio (Naive MV) and the Robust Ordinal portfolio. Both portfolios are subject to long-only and budget constraints ($w \geq 0, w^\top \mathbf{1} = 1$). Panel A reports the Herfindahl Index, a measure of concentration where a value of 1.0 indicates a single-asset portfolio. Panel B reports the cross-sectional standard deviation of the portfolio weights. Panel C reports the number of active positions (assets with strictly positive weights) in the portfolio.

9 Conclusion

This paper proposes a robust optimization framework that addresses a core fragility of mean-variance portfolio selection: its extreme sensitivity to cardinal estimation errors. By defining uncertainty in purely ordinal terms—preserving only the ranking and sign of expected returns—we show that it is possible to construct portfolios that are structurally immune to the failures of standard plug-in optimization.

The key theoretical contribution is the formalization of an ordinal uncertainty set and the proof that optimizing against its worst-case realization yields a bounded-regret portfolio. We show that this problem can be reformulated as a tractable convex quadratic program using duality, proving that robustness does not come at the cost of computational complexity. This "ordinal regularization" offers a foundation for ranking-based investment strategies, which have long outperformed despite

discarding precise return estimates.

Our numerical simulations and empirical results validate these theoretical insights. In controlled simulations, we demonstrate that while the regret of the standard mean–variance optimizer explodes quadratically with cardinal error, the robust ordinal portfolio maintains a strictly bounded utility loss. Empirically, applied to a universe of U.S. equities sorted by analyst recommendations, the robust framework produces portfolios with significantly greater breadth, lower turnover, and superior out-of-sample performance compared to the plug-in benchmark. These findings highlight that discarding unreliable magnitudes is not a weakness, but a principled and effective response to pervasive estimation risk.

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